

AI Mini-project

PartFlow

• MES

LOT • , AI MES

{ } • { }

2026.09

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PartFlow

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Manufacturing Execution System

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OpenAPI ・

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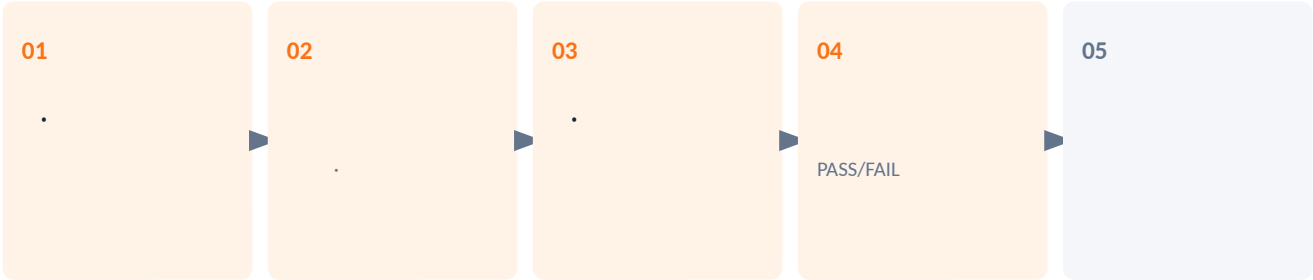
MES ?

AI

PartFlow

MES (LOT ・ ・AI) ,

MES

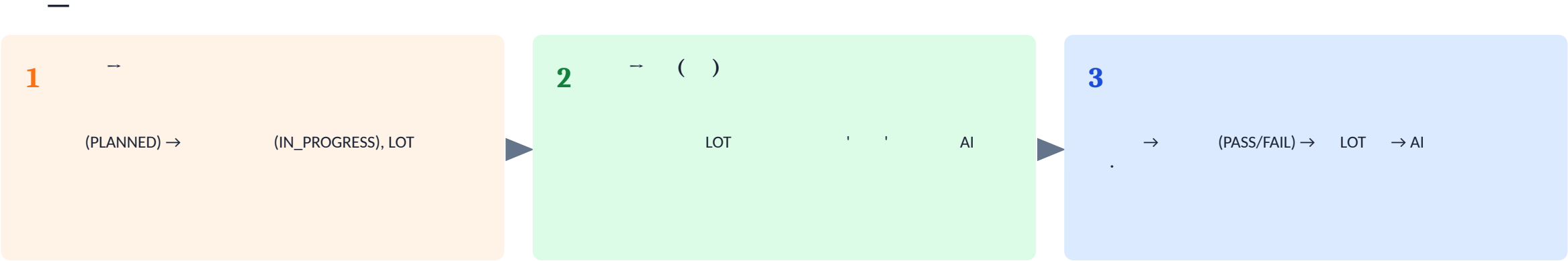
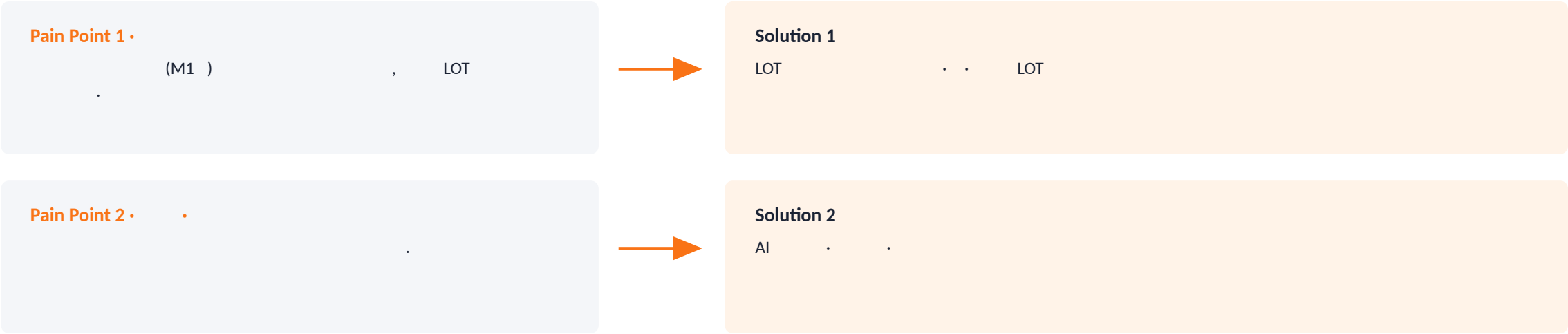


PartFlow 01~04(・) , 05 .

AI

MES ・ ・ , AI .

LOT



· · 3 + AI

· ,

, LOT ·

, LOT , AI ·

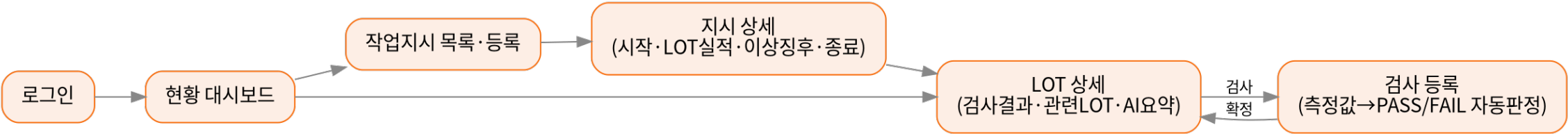
AI

Spring AI

(/)

F03(·) () () .

→ → → LOT →

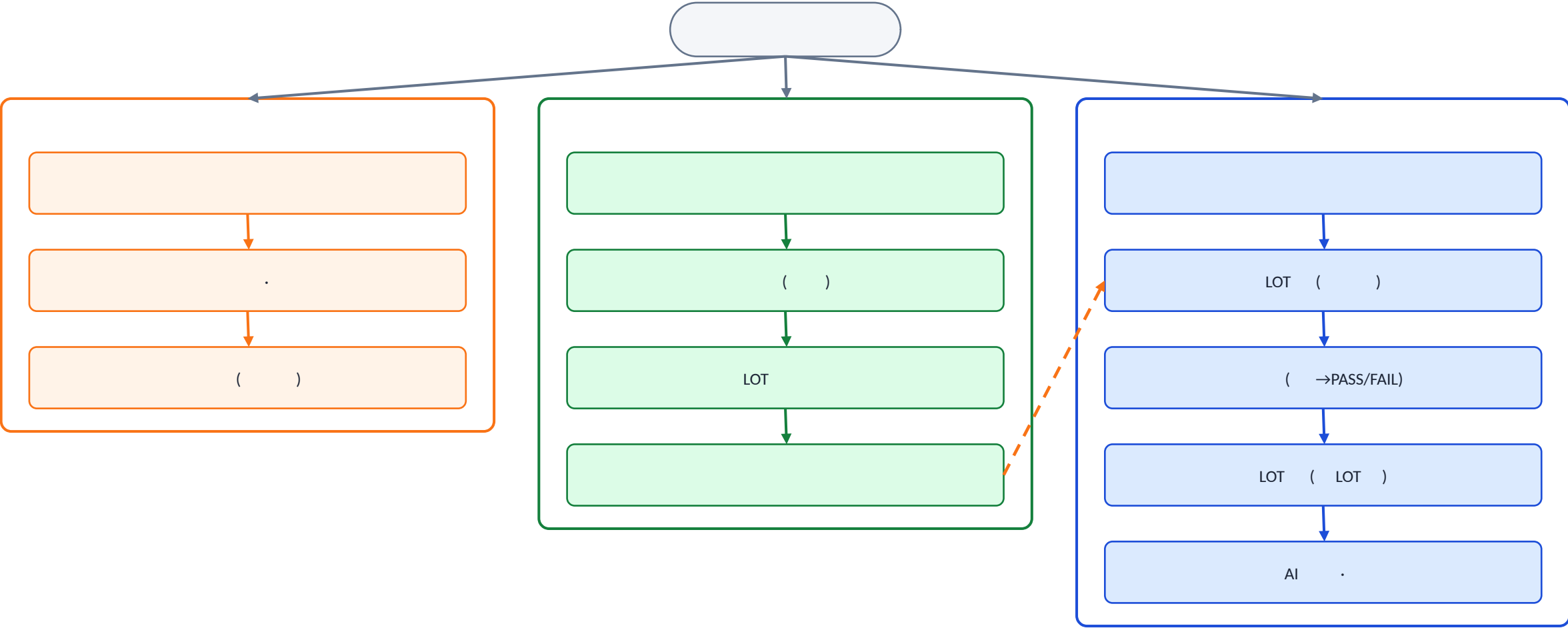


S01	S02	S03	S04	S05	S06	+
		.		LOT		.

(S02) , .

F02	F04 LOT	F05	F09 LOT	F07/F08 AI .
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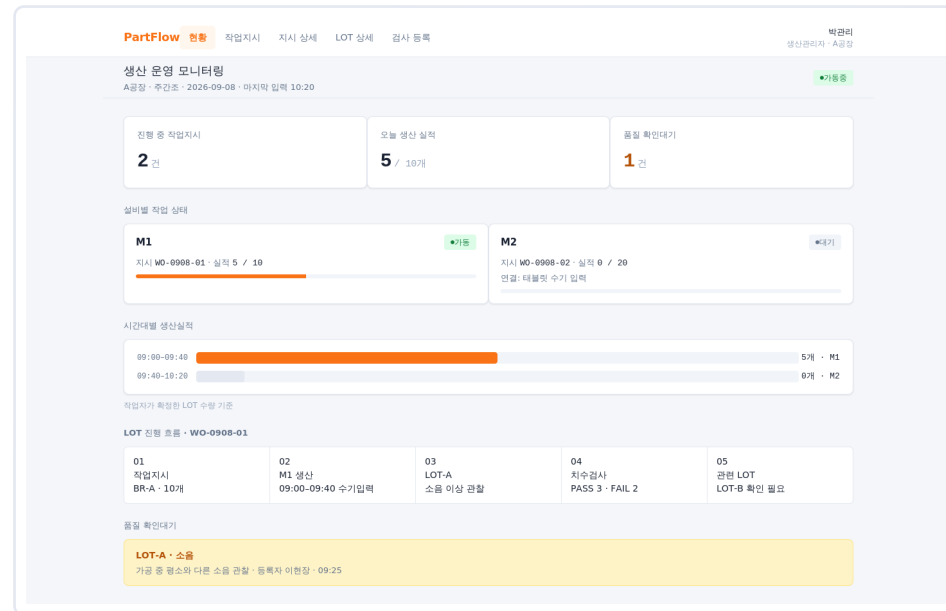
— · · (PPT)





S01

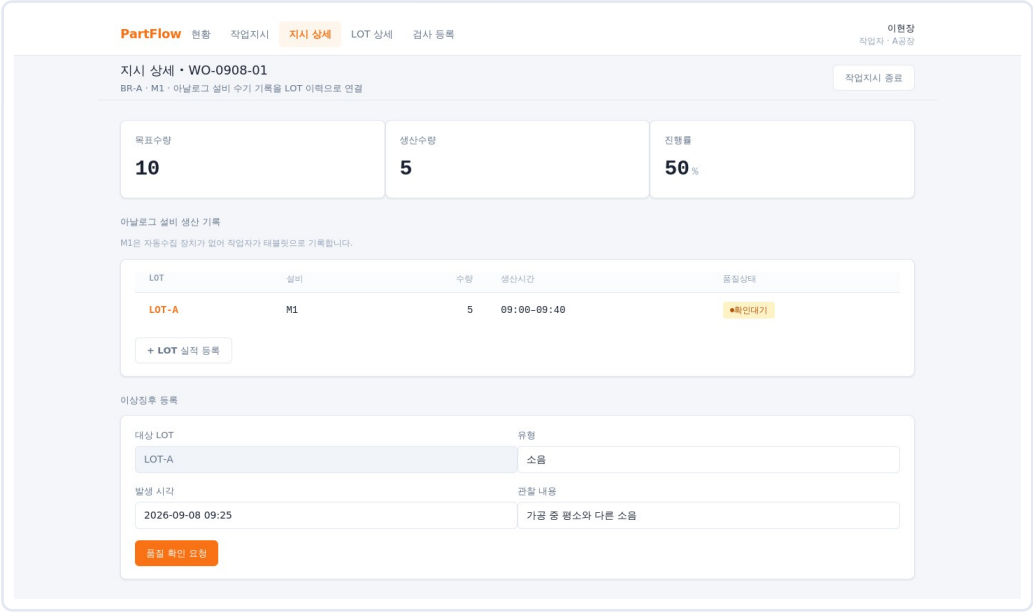
- () 3
- POST /auth/login



S02

- ,
- GET /dashboard

S03 · · S04



S03 ·

- / ,
- GET·POST /work-orders

S04

- ·LOT ·
- POST /work-orders/{id}/lots

S05 LOT · S06 ()

PartFlow현황작업지시지시 상세LOT 상세검사 등록

김동길
품질담당자 · A공정

LOT 상세 · LOT-A
WO-0908-01 · BR-A

•검사완료 · FAIL 2

생산 · 규격

설비M1

생산시간09:00-09:40

규격 목표값10.00 mm

허용 범위 (구멍 지름)9.90-10.10 mm

검사 결과

PASS 3 · FAIL 2

#1 9.90#2 10.00#3 10.10#4 9.89#5 10.11

관련 LOT 후보

LOT	생산시간	검사	신장 근거
☒ LOT-B	09:40-10:20	•대기	같은 BR-A · M1 · 같은 생산일

선택 메모 AI 요약 생성

AI 요약 초안

AI 초안 · LOT-A 규격 이탈 2건(#4, #5). 원인은 기록상 확인되지 않음. 같은 시간대 M1에서 생산된 LOT-B는 아직 검사되지 않음 — 검사 진행 여부를 확인해 주세요.

S05 LOT

- LOT + AI
- GET /lots/{id}, /lots/{id}/related

PartFlow현황작업지시지시 상세LOT 상세검사 등록

김동길
품질담당자 · A공정

검사 등록 · LOT-A
생산수량 5개(원기전용) · 허용 규격 9.90-10.10 mm

측정값 입력

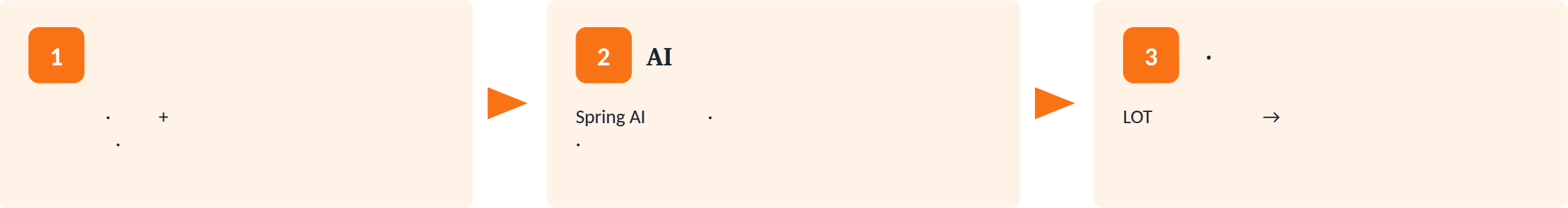
제품 순번	측정값 (mm)	예상 판정
1	9.90	•PASS
2	10.00	•PASS
3	10.10	•PASS
4	9.89	•FAIL
5	10.11	•FAIL

검사 메모
하한/상한 이탈 각 1개, 원인은 미확인

검사 확정

S06

- → PASS/FAIL
- POST /lots/{id}/inspection



AI

×

× /

×

(·)

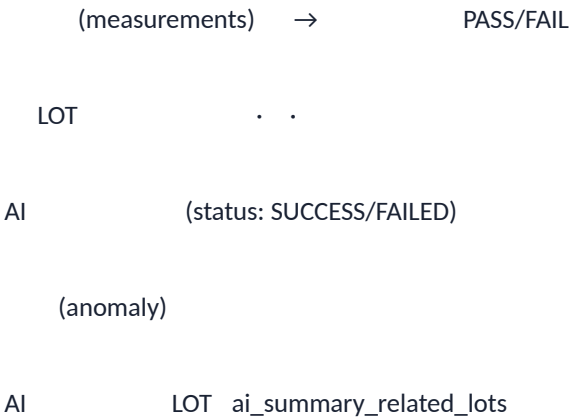
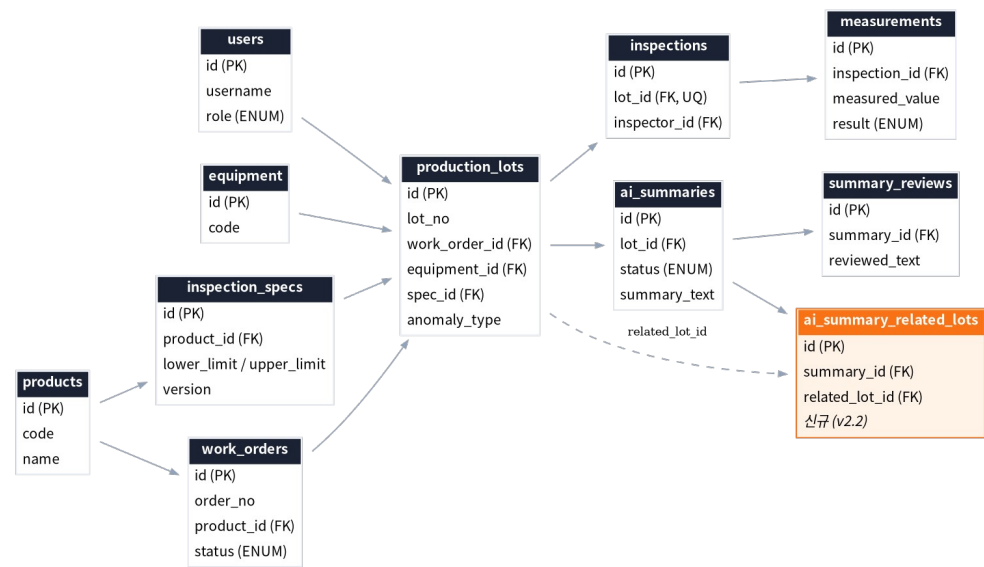
(status: FAILED)

— AI

AI (summary_reviews)

“LOT-A 2 (#4, #5). M1 LOT-B — .”

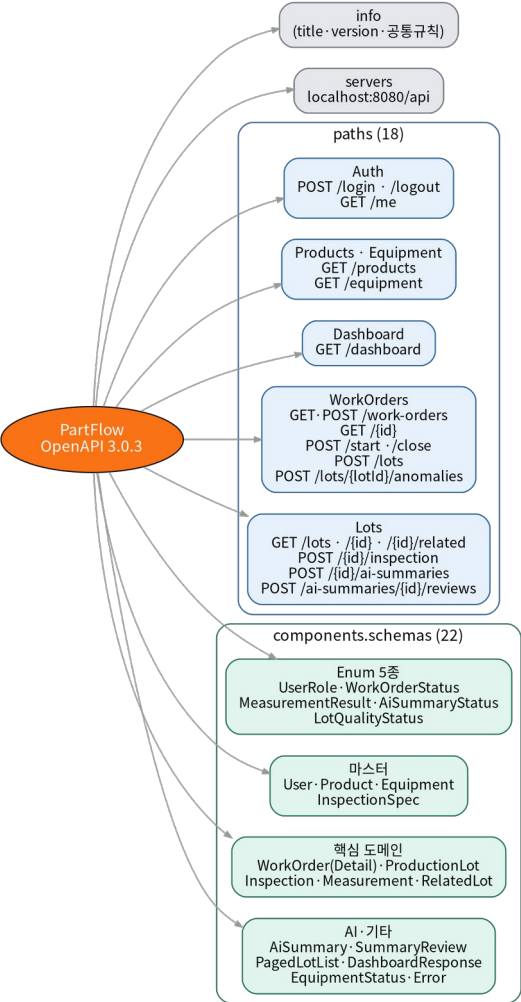
ERD ·



ENUM

Enum	
user_role	MANAGER / OPERATOR / QUALITY
work_order_status	PLANNED / IN_PROGRESS / CLOSED
measurement_result	PASS / FAIL
ai_summary_status	SUCCESS / FAILED

OpenAPI 3.0.3 · 18 paths · 22 schemas · Security Schemes



- POST

/work-orders/{id}/lots
LOT
- POST

/lots/{id}/inspection
→ PASS/FAIL
- GET

/lots/{id}/related
LOT
- POST

/lots/{id}/ai-summaries
LOT (relatedLotIds) AI ,
- GET

/dashboard
· ·

: POST /auth/login , Error (\$ref) .

+ HTTP +

HTTP

400		,
401		
403		API
404		·LOT·
409	·	,
502/504	AI /	,

	·
	400 fieldErrors
	401
·	502/504

```
{ "code", "message", "fieldErrors": [{ "field", "reason" }] } —
```

Error \$ref

1

{ _ _PartFlow}- .pdf

· · ·UI · 8

2

API

{ _ _PartFlow}-API.yml

OpenAPI 3.0 · 18 paths · 22 schemas

3

DB

{ _ _PartFlow}-DB.dbml

dbdiagram.io

Swagger UI API ()

· , (BR-A/BR-B). · · · · ·

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1

・ ・API・ ,
・ ・

2

AI (: LOT ,) , MES
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